

Flooding and Drying in Discontinuous Galerkin Finite-Element Discretizations of Shallow-Water Equations. Part 2: Two Dimensions.

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Abstract

“Flooding” and “drying” in the shallow-water equations consists of the time-dependent motion of the water line or free boundary over topography. A novel numerical algorithm is developed to deal with flooding and drying in a second-order space discontinuous finite-element method. The mesh has fixed Eulerian elements and mixed Eulerian-Lagrangian elements at free boundaries. The algorithm contains the following aspects. The weak formulation is adapted to include the movement of the nodes on the free boundary. A time-step restriction is derived by analyzing the HLLC numerical flux used in order to ensure positivity of the mean water depth. The mesh quality is maintained by updating the mesh topology by locally deforming an underlying basic mesh of rectangles. A good triangular mesh is found by locally swapping or inserting diagonals in this (locally deformed) rectangular mesh. The water depth and the velocity are approximated using mean and slope information. Finally, even although the mean water depth is ensured to be positive, the water depth is adjusted when it falls below zero by limiting the slope information of the depth. The algorithm devised can also be applied or extended to other mathematical models with a two-dimensional continuity equation, such as the two-dimensional compressible equations of motion.

Keywords: shallow-water equations, flooding and drying, free-boundary dynamics, discontinuous finite-element method, two-dimensional mesh topology, positivity of mean water depth

AMS Subject Classification: 76M10, 65M60

1 Introduction

We consider explicit space discontinuous finite- element discretizations of the free boundary dynamics in the two-dimensional shallow-water equations, in extension of the one-dimensional algorithm of Bokhove and Wirosoetisno (2003), and Bokhove (2003). Shallow-water equations are hyperbolic in their conservative limit, and describe the approximate dynamics of the flow averaged over the depth of a thin layer of water. The domain of the shallow-water equations is defined by the area where the depth h is positive. In many practical applications, part of the boundary of the domain is defined by a free boundary where the layer thickness h reaches zero, while other parts of the boundary are impenetrable walls, for example. The movement of this free boundary or water line over topography results into “flooding” and “drying”. In a bounded two-dimensional area (or computational domain) (partly) filled with fluid, either free boundaries are absent, or each free boundary is a closed loop within the bounded area or begins and ends at the boundary of this area. Flooding occurs when the fluid inundates dry area: the topology of a free boundary then either remains the same or changes when points of (disconnected pieces of) free boundary meet, which may change the topology of the free boundary. Drying occurs when the fluid retracts from wetted areas: it arises either within the existing free-boundary topology, or with the creation of a new free boundary when a dry area emerges within a previously wet area. The design of a discontinuous finite-element approximation for shallow-water dynamics with flooding and drying in two horizontal dimensions is the topic of the present paper.

The advantages of discontinuous (Galerkin) finite-element methods are as follows (Cockburn *et al.*, 1989; Cockburn and Shu, 1998):

- i. It is straightforward to vary the order of accuracy per element (so-called p -adaptivity).
- ii. The use of unstructured meshes is relatively straightforward, as well as local mesh refinement (so-called h -adaptivity).
- iii. The heart of the method concerns the communication transfer via the fluxes through the elemental faces. Hence, the method is extremely local which is used in the present paper to deal with the free-boundary dynamics in the shallow-water equations. Such localness allows also for efficient parallelization, and the accurate implementation of other boundary conditions such as in- and outflow conditions.

However, the overall formulation is more complex, and more degrees of freedom are involved relative to finite-volume or finite-difference schemes.

The free-boundary algorithm developed consists of the following steps: a) the inclusion of time dependence in the weak formulation and its (explicit) discretization for the mixed Eulerian-Lagrangian free-boundary elements next to the fixed Eulerian flooded elements; b) the use of a HLLC flux through fixed and time-dependent faces using the approach by Toro *et al.* (1994) and extended to the moving case by Van der Vegt and Van der Ven (2002); c) the use of a time-step criterion to ensure the positivity of the mean depth in each element; and d) a treatment of the evolving (triangular) mesh within the fixed topology of an underlying regular mesh of rectangles by allowing edge swapping of diagonals within these rectangles, and local mesh deformation and upgrading after each time step. The above steps are similar to the one-dimensional case developed in Bokhove and Wirosoetisno (2003) and Bokhove (2003), except for the added complexity in the regular upgrading of the mesh topology which appears to be a distinct research area in its own right. As a consequence, the mesh upgrading proposed hinges around using a basic regular mesh of rectangles. Nevertheless, the approach becomes partially unstructured as this regular grid is deformed around curved or piecewise linear curves of fixed or free boundaries. It is unknown whether such mesh upgrading can be done successfully for a fully unstructured triangular mesh. Closely related research in computational geometry concerns a Delaunay approach to cutting in triangulated surfaces (Nienhuys, 2003). The difference is that cutting requires mesh adaptation around the incision point only, while the movement of the free boundary requires adaptation along the whole water line in the shallow-water equations.

The plan of the paper is unfolded next. The shallow-water equations are introduced in §2. A

space discontinuous finite-element method is formulated for elements with moving nodes in §3. The spatial flux through the faces in the finite-element formulation used is the HLLC flux, introduced in §4. Both Eulerian interior elements and mixed Eulerian-Lagrangian free-boundary elements are included. The discretization in these free-boundary elements is explained in §5. The time-dependent topology of the triangulation, the heart of the algorithm, is explained in §5.2. Analysis of the HLLC flux combined with the free-boundary treatment for a given time discretization shows that a time-step restriction guarantees that the mean depth in an element remains positive. In combination with a restriction on the slope such that the depth is set to zero when it becomes negative (in essence an ad hoc slope limiter in regions where dry patches emerge), a robust and reasonably accurate method is developed that is designed to simultaneously deal with flooding and drying, and discontinuities such as shocks, hydraulic bores and jumps. Conclusions are drawn in §6.

2 Shallow-Water Equations

The shallow-water equations are (*e.g.* Pedlosky, 1987)

$$\begin{aligned}\partial_t h + \partial_x(h u) + \partial_y(h v) &= 0 \\ \partial_t(h u) + \partial_x(h u^2 + P) + \partial_y(h u v) &= S_2, \\ \partial_t(h v) + \partial_x(h u v) + \partial_y(h v^2 + P) &= S_3;\end{aligned}\tag{2.1}$$

here the horizontal velocity field $(u, v) = (u(\bar{x}, t), v(\bar{x}, t))$, $h(\bar{x}, t)$ is the depth of the layer as function of the horizontal coordinates $\bar{x} = (x, y)^T$ and time t , and the effective “pressure” or potential is $P = P(h) = g h^2/2$ with g the gravitational acceleration. The terms $S_{2,3}$ depend in general on x, y and t explicitly, and implicitly on the variables u, v, h . In particular, for a rotating fluid over topography at $z = h_B(\bar{x})$ these terms have the form $S_2 = f h v - h \partial_x h_B$, $S_3 = -f h u - h \partial_y h_B$ with Coriolis parameter $f = f(x, y)$. In addition, partial derivatives are denoted by $\partial_t = \partial/\partial t$, etcetera.

To facilitate the numerical formulation, we rewrite (2.1) concisely as

$$\partial_t U + \operatorname{div} F(h, u, v) = S\tag{2.2}$$

with $(\operatorname{div} F)_i = \partial_{x_j} F_{ij}$ and $\partial_{x_j} = \partial/\partial x_j$, where $i = 1, 2, 3$ and $j = 1, 2$; state vector $U = (h, h u, h v)^T$; flux tensor

$$F(h, u, v) = \begin{pmatrix} h u & h v \\ h u^2 + P & h u v \\ h u v & h v^2 + P \end{pmatrix};\tag{2.3}$$

and vector $S = (0, S_2, S_3)^T$ of external contributions.

The flow domain $\Omega_f \in \mathcal{R}$ may be time dependent and has a boundary $\partial\Omega_f$. It is embedded within a fixed computational domain Ω with boundary $\partial\Omega$. At the fixed boundary segments, the boundary conditions specify the in- and outflow, or no through flow at walls. The free boundary is in general specified by $h(\bar{x}, t) = 0$. The system (2.1) or (2.2) is completed with initial conditions $U_0 = U(\bar{x}, 0)$.

3 Space Discontinuous Finite-Element Discretization

3.1 Finite-element topology of flooded and dry areas

The (computational) domain Ω is a bounded area in $\mathbb{R}^2$ with fixed boundary $\partial\Omega$. Consider the open spatial flow domain $\Omega_f \in \Omega$ with boundary $\partial\Omega_f$. The boundary $\partial\Omega_f$ consist of (parts of) the boundary of Ω , denoted by $\partial\Omega_p$, and a free boundary, denoted by $\partial\Omega_{fb}$, such that $\partial\Omega_f = \partial\Omega_p \cup \partial\Omega_{fb}$. The dry part Ω_d of Ω is such that $\Omega_f \cup \Omega_d = \Omega$ and has boundary $\partial\Omega_d$.

Introduce a tessellation $\mathcal{T}_{sh}$ of N_{el}^s open fixed or solid elements in the domain Ω

$$\mathcal{T}_{sh} = \{K_k^s | \bigcup_{k=1}^{N_{el}} \bar{K}_k^s = \bar{\Omega} \text{ and } K_k^s \cap K_{k'}^s = \emptyset \text{ if } k \neq k', 1 \leq k, k' \leq N_{el}^s\} \quad (3.1)$$

with $\bar{K}_k$ the closure of K_k and so forth.

Definition i. A path in the fixed triangulation $\mathcal{T}_{sh}$ consist of a sequence of connected faces of the triangles between the nodes. A path is either closed, when it begins or ends at the same node, or open, when it begins at a boundary node and ends at a different boundary node. When there are multiple paths in Ω they are not allowed to intersect and not allowed to share two nodes of a triangle.

Definition ii. Triangular elements along a path consist of Type 0, Type I, or Type II, elements in which zero nodes, one node or two nodes of the triangle do not coincide with the nodes of the path, respectively.

Definition iii. A path in the fixed tessellation of triangles is allowable when elements on either side along the path consist only of Type I and II elements.

Definition iv. An allowable path or discretized free boundary L_f in the fixed tessellation $\mathcal{T}_{sh}$ is optimal if a property such as the area between L_f and a given free boundary $\partial\Omega_{fb}'$ in Ω is minimized.

It appears that each free boundary $\partial\Omega_{fb}'$ in Ω has at least one optimal path L_f .

Definition v. Define tessellation $\mathcal{T}_h$ and $\mathcal{T}_{dh}$ for the flooded and dry areas Ω_f and Ω_d such that: a) the tessellation $\mathcal{T}_h$ consists of N_{el} open elements K_k in Ω_f

$$\mathcal{T}_h = \{K_k | \bigcup_{k=1}^{N_{el}} \bar{K}_k = \bar{\Omega}_f \text{ and } K_k \cap K_{k'} = \emptyset \text{ if } k \neq k', 1 \leq k, k' \leq N_{el}\}, \quad (3.2)$$

where K_k depends on time for the moving nodes on $\partial\Omega_f$. b) In Ω_d there is a tessellation $\mathcal{T}_{dh}$ with $N_{el}^d = N_{el}^s - N_{el}$ such that the free boundary nodes on $\partial\Omega_{fb}$ in $\mathcal{T}_{dh}$ correspond to the free boundary nodes on $\partial\Omega_{fb}$ in $\mathcal{T}_h$. Hence, the tessellations $\mathcal{T}_h + \mathcal{T}_{dh}$ form a tessellation $\mathcal{T}_{rh}$ of Ω . c) The path L_{fb} of boundary nodes on $\partial\Omega_{fb}$ is allowable in $\mathcal{T}_{rh}$. c) The tessellations $\mathcal{T}_{rh}$ and $\mathcal{T}_{sh}$ are topologically equivalent. d) The nodes of the tessellations $\mathcal{T}_{rh}$ and $\mathcal{T}_{sh}$ coincide except for the nodes on the free boundary $\partial\Omega_{fb}$ and a discretized free boundary L_f of $\mathcal{T}_{sh}$ such that L_f is an optimal allowable path of L_{fb} .

The geometry changes below are easier to visualize when L_{fb} and L_f coincide at the initial time. At time t^n , path L_f is an optimal allowable path of L_{fb}^n and the tessellation $\mathcal{T}_h$ is topologically equivalent with “flooded” elements in $\mathcal{T}_{sh}$. The evolution of K_k in a time step $t \in (t^n, t^{n+1})$ is (then) such that topology of the tessellation $\mathcal{T}_h$ is preserved. However, at time $t \downarrow t^{n+1}$ the free boundary nodes on L_{fb} will have moved and do no longer coincide with L_f . Furthermore, L_f may no longer be the optimal path. The question is then how we efficiently choose this optimal path in $\mathcal{T}_{sh}$ at time t^{n+1} . In §5.2, this question is answered for a triangulation $\mathcal{T}_{sh}$ based on a mesh of rectangles locally deformed to conform to the domain Ω .

Finite-element discretizations of (2.2) are considered with approximations U_h and w_h to the state vector U and test functions w , to be introduced shortly, respectively. These are such that U_h and w_h belong to the broken space

$$V_h = \{w : w|_{K_k} \in P^{d_P}(K_k), k = 1, \dots, N_{el}\}, \quad (3.3)$$

in which $P^{d_P}(K_k)$ denotes the space of polynomials in K_k of degree $d_P = d_P(U)$. Test functions w_h can be time dependent.

3.2 Weak formulation

We multiply (2.2) by test functions $w_h = w_h(\bar{x}, t)$ and use Gauss' theorem in space to rewrite the flux term. After summing over all elements the following weak formulation emerges

$$\sum_K \left\{ \int_{K_k} w_h \cdot \partial_t U \, dK + \int_{\partial K_k} \hat{n}_k^- \cdot ((w_h^-)^T F(U^-)) \, d\Gamma - \int_{K_k} (\text{grad } w_h)^T : F(U) \, dK - \int_{K_k} w_h \cdot S(U) \, dK \right\} = 0, \quad (3.4)$$

where $(\text{grad } w_h)_{ji} = (\text{grad } w_h)_{ij}^T = \partial_{x_j} w_{hi}$; the diadic product $(\text{grad } w_h)^T : F = (\partial_{x_j} w_{hi}) F_{ij}$ ($i = 1, 2, 3$ and $j = 1, 2$); dK is an infinitesimal area of element K ; ∂K is the boundary of element K ; $\hat{n}_k^-$ is the outward normal of the element K_k with infinitesimal boundary segment $d\Gamma$; and where we used the trace relations $\gamma^\pm(U) = \lim_{\epsilon \downarrow 0} U(\bar{x} \pm \epsilon \hat{n})$. In index notation, Gauss' rule is used after writing

$$w_{hi} \partial_{x_j} F_{ij} = \partial_{x_j} (w_{hi} F_{ij}) - F_{ij} \partial_{x_j} w_{hi}. \quad (3.5)$$

The communication in the shallow-water system between elements should occur by mass and momentum transfer due to the flux term F at the element faces. But in the weak formulation (3.4) there is an ambiguity since w_h is discontinuous. Each element lies isolated in space in the discretization (3.16). Fluxes at either side of a face are therefore not continuous at this stage.

Given an ordering of the faces, we distinguish the elements k_l and k_r to the left and right of a face. Consider the integrals over the element boundaries with an ordering such that we have outward normals and limiting trace values per face for a left element with $\hat{n}^l, F^l$ and U_l , and a right side with $\hat{n}^r, F^r$ and U_r , as follows

$$\begin{aligned} \sum_{K_k} \int_{\partial K_k} \hat{n}_k^- \cdot ((w_h^-)^T F^-) \, d\Gamma &= \sum_S \int_S \hat{n}^l \cdot ((w_h^l)^T F^l) + \hat{n}^r \cdot ((w_h^r)^T F^r) \, d\Gamma \\ &= \sum_S \int_S (\hat{n}^l \cdot (w_h^l)^T + \hat{n}^r \cdot (w_h^r)^T) (\alpha F^l + \gamma F^r) + \\ &\quad (\gamma (w_h^l)^T + \alpha (w_h^r)^T) \cdot (\hat{n}^l F^l + \hat{n}^r F^r) \, d\Gamma. \end{aligned} \quad (3.6)$$

with $\alpha + \gamma = 1$ and $\alpha, \gamma \geq 0$. A counterclockwise convention is used, which implies that there is no element right of a boundary face. The last two terms in (3.6) are zero if the flux is taken to be conservative

$$\int_S \hat{n}^l \cdot ((w_h)^T F^l) \, d\Gamma = - \int_S \hat{n}^r \cdot ((w_h)^T F^r) \, d\Gamma \quad (3.7)$$

with w_h evaluated on either side of a face and $\hat{n}^l = -\hat{n}^r$ at a face. Consequently, we are left with

$$\begin{aligned} \sum_{K_k} \int_{\partial K_k} \hat{n}_k^- \cdot ((w_h^-)^T F^-) \, d\Gamma &= \sum_S \int_S \hat{n}^l \cdot ((w_h^l)^T (\alpha F^l + \gamma F^r) - (w_h^r)^T (\alpha F^l + \gamma F^r)) \, d\Gamma \\ &= \sum_{K_k} \int_{\partial K_k} \hat{n}^- \cdot ((w_h^-)^T (\alpha F^- + \gamma F^+)) \, d\Gamma, \end{aligned} \quad (3.8)$$

in which the flux is a linear combination of the flux on either side of the face.

Sofar, we have used U and not its approximation U_h , which allowed us to use continuity of the flux in (3.7). For numerical purposes, we approximate U across elements in a discontinuous manner by U_h which is continuous in each element K_k . Subsequently, we replace the flux $\alpha F^l + \gamma F^r$ in the flux term (3.8) of the weak formulation (3.4) per element by a continuous numerical flux $\tilde{F}(U_{hl}, U_{hr}, \hat{n})$ or $\tilde{F}(U_h^-, U_h^+, \hat{n})$, which depends on the discontinuous values U_{hl}, U_{hr} or $U_h^\pm$ directly adjacent to each interface. When no confusion arises we simply write $U_{l,r}$ and $U^\pm$ for the discontinuous

variables U_{hl}, U_{hr} and $U_h^\pm$ in element K_k . Hence, the numerical flux $\tilde{F}(U_l, U_r, \hat{n})$ replaces F^l and F^r at that face and thus guarantees conservation. The numerical flux must satisfy the following conditions (Shu and Osher, 1989): i) $\tilde{F}$ is consistent, *i.e.*, $\tilde{F}(U, U) = F(U)$; and ii) $\tilde{F}(U, V)$ is a non-decreasing function in both of its arguments; and, iii) $\tilde{F}(U, V)$ must be conservative, *i.e.*, $\hat{n}_K \cdot \tilde{F}(U, V) = -\hat{n}_K \cdot \tilde{F}(V, U)$.

By replacing U by U_h and the flux F by its numerical counterpart $\tilde{F}$, we arrive at the following discretized weak formulation

$$\sum_K \int_{K_k} w_h \cdot \partial_t U_h - (\text{grad } w_h)^T : F(U_h) - w_h \cdot S(U_h) \, dK + \int_{\partial K_k} \hat{n}_k^- \cdot ((w_h^-)^T \tilde{F}(U^-, U^+, \hat{n})) \, d\Gamma = 0 \quad (3.9)$$

or

$$\begin{aligned} \sum_K \int_{K_k} w_h \cdot \partial_t U_h - (\text{grad } w_h)^T : F(U_h) - w_h \cdot S(U_h) \, dK + \\ \sum_S \int_S \hat{n}^l \cdot ((w_h^l)^T - (w_h^r)^T) \tilde{F}(U_l, U_r, \hat{n}) \, d\Gamma = 0. \end{aligned} \quad (3.10)$$

The latter form emphasizes the numerical strategy by separately calculating the fluxes in a sum over all faces and updating the continuous flux in the two neighboring elements per face, and subsequently the element integrals in a sum over all elements.

3.3 Polynomial basis functions

The domain $\Omega = \Omega_f \cup \Omega_d \subset \mathbf{R}^2$ is partitioned into triangular elements. It is convenient to introduce a reference element $\hat{K}$ and define the mapping $F_K : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ between the reference element $\hat{K}$ and element K_k as follows

$$\bar{x} = F_{K_k}(\bar{\zeta}) = \sum_{m=0}^{N_{\text{el}}-1} \bar{x}_{k,m} \chi_m(\bar{\zeta}) \quad (3.11)$$

with $\bar{\zeta} = (\zeta_1, \zeta_2)^T$. Triangles have $N_{\text{el}} = 3$ nodes $\bar{x}_{k,m}$, corresponding to nodes $(0,0)^T, (1,0)^T, (0,1)^T$ in the reference element, and

$$\chi_0(\bar{\zeta}) = \zeta_0 = 1 - \zeta_1 - \zeta_2, \quad \chi_1(\bar{\zeta}) = \zeta_1, \quad \chi_2(\bar{\zeta}) = \zeta_2. \quad (3.12)$$

In each element, we approximate U_h and w_h in a polynomial expansion

$$U_h(\bar{x}, t) = \bar{U}_k + \sum_{m=1}^M \hat{U}_{k,m} \psi_{m,k}(\bar{x}, t) \quad \text{and} \quad w_h(\bar{x}, t) = \bar{W}_k + \sum_{m=1}^M \hat{W}_{k,m} \psi_{m,k}(\bar{x}, t). \quad (3.13)$$

The basis functions are polynomials in ζ_i of order $d_P = d_P(U)$ chosen such that

$$\psi_{m,k}(\bar{x}, t) = \begin{cases} 1 & \text{if } m = 0 \\ \varphi_{m,k}(\bar{x}, t) - c_{m,k} & \text{if } m \geq 1 \end{cases} \quad (3.14)$$

with

$$c_{m,k} = \int_{K_k} \varphi_{m,k}(\bar{x}, t) \, d\bar{x} / |K_k|, \quad (3.15)$$

$\varphi_{m,k} = \zeta_1^{k_m-l} \zeta_2^l$ for $l = 0, \dots, k_m$, $k_m \leq d_P$, $\varphi_0 = 1$, an appropriate integer M , and $|K_k|$ the area of an element. Hence, $\int_{K_k} \psi_{m,k} \, d\bar{x} = 0$ for $m > 0$.

3.4 Discretized weak formulation

After substituting (3.13) into (3.9), we obtain in each element the following equations for the mean and fluctuating parts, since $\bar{W}_k$ and $\hat{W}_{k,m}$ for $m = 1, \dots, M$ are arbitrary,

$$\begin{aligned} |K_k| \frac{d}{dt} \bar{U}_k + \hat{U}_{k,n} \int_{K_k} \partial_t \psi_{n,k} dK - \int_{K_k} S dK + \int_{\partial K_k} \tilde{F} \cdot \hat{n}_k d\Gamma = 0 \\ \frac{d\hat{U}_{k,n}}{dt} \int_{K_k} \psi_{m,k} \psi_{n,k} dK + \hat{U}_{k,n} \int_{K_k} \psi_{m,k} \partial_t \psi_{n,k} dK - \int_{K_k} \psi_{m,k} S dK - \int_{K_k} F \cdot \text{grad } \psi_{m,k} dK + \\ \hat{U}_{k,n} \int_{K_k} (\text{grad } \psi_{m,k})^T \cdot D(U_h) : (\text{grad } \psi_{n,k})^T dK + \int_{\partial K_k} \psi_{m,k}^- \tilde{F} \cdot \hat{n}_k d\Gamma = 0, \end{aligned} \quad (3.16)$$

where we also introduced a friction term with dissipation matrix operator $D(U_h) > 0$ intended to stabilize the integration of $\hat{U}_{k,n}$ with $m, n = 1, \dots, M$. Note that

$$(\text{grad } \psi_{m,k})^T \cdot D(U_h) : (\text{grad } \psi_{n,k})^T = \partial_{x_i} \psi_{m,k} D_{ij} \partial_{x_j} \psi_{n,k}. \quad (3.17)$$

The third-order Runge-Kutta method of Shu and Osher (1989), for example is used to discretize (3.16) in time. In Appendix A, the boundary and interior integrals in (3.16) are given in detail for triangular elements. The equations then become

$$\begin{aligned} \frac{d}{dt} (|K_k| \bar{U}_k) - \int_{K_k} S dK + \int_{\partial K_k} \tilde{F}_V d\Gamma = 0 \\ \frac{d\hat{U}_{k,n}}{dt} \int_{K_k} \psi_{m,k} \psi_{n,k} dK + \int_{K_k} \partial_{x_i} (V_i \psi_m) U_k dK + \hat{U}_{k,n} \int_{K_k} \psi_{m,k} \partial_t \psi_{n,k} dK - \int_{K_k} \psi_{m,k} S dK - \\ \int_{K_k} F \cdot \text{grad } \psi_{m,k} dK + \hat{U}_{k,n} \int_{K_k} (\text{grad } \psi_{m,k})^T \cdot D(U_h) : (\text{grad } \psi_{n,k})^T dK + \int_{\partial K_k} \psi_{m,k}^- \tilde{F}_V d\Gamma = 0 \end{aligned} \quad (3.18)$$

with the numerical flux $\tilde{F}_V$ replacing $F_V = F \cdot \hat{n}_k - V U$ and $V = \mathbf{V} \cdot \hat{n}_k$ the normal velocity of a moving triangular element (at a face), and V_i its velocity components.

4 HLLC Flux

Since U is discontinuous across element faces, we need to approximate the flux $F_V = F \cdot \hat{n}_k - V U$ by a numerical flux $\tilde{F}_V$. Approximating the variable U adjacent to a face at a certain time t_n locally by constant values to the left and right, denoted by U_l and U_r , respectively, one obtains a Riemann problem which can be solved exactly. The resulting Riemann flux, however, is expensive to compute. An accurate and efficient alternative is the HLLC flux of Toro *et al.* (1994) for hyperbolic systems. In this section, we summarize the HLLC flux.

The choice of numerical flux should ensure positivity of the mean fluid depth under a certain time-step restriction—given an explicit time step routine. It is undesirable to choose a numerical method in which the depth stays positive everywhere, since the physics allows dry patches to emerge. If the depth is approximated using (3.13) by a mean and fluctuating parts, information contained in the coefficients $\hat{U}_{k,m}$ ($m > 0$) can be used to signal when and where the dry regions emerge, that is, where the water depth becomes zero. In §5.3, we show that the HLLC flux approximation for interior non-moving elements and moving free-boundary elements combined with an Euler forward time discretization and a maximum principle ensures that the mean depth in each element remains positive.

In the derivation of the HLLC flux, the component of the flux normal to a face is considered in a one-dimensional Riemann problem. Define the outward unit normal of a face in element K_k as $\hat{n}_k =$

$(n_1, n_2)^T$ the tangent unit vector $\boldsymbol{\tau} = (-n_2, n_1)$, and a rotated coordinate system with coordinates $\hat{x}$ in the normal direction and $\hat{y}$ in the tangent direction. Ignoring the external contribution in the term S , the weak formulation of (2.2) in $\hat{x}, \hat{y}$ -coordinates along the respective face can be written as

$$\int \int (U d\hat{x} + F \cdot \hat{n}_k dt) d\hat{y} = 0. \quad (4.1)$$

Hence, consider the one-dimensional Riemann problem

$$\partial_t U + \partial_{\hat{x}} \hat{F}(U) = 0 \quad (4.2)$$

with $U = (h, h u, h v)^T$, $\partial_{\hat{x}} = \hat{n}_k \cdot (\partial_x, \partial_y)^T$, $\hat{F}(U) = F(U) \cdot \hat{n}_k = (h q, h q(u, v) + \hat{n}_k^T P(h))^T$ and $q = \hat{n}_k \cdot (u, v)^T$, and the constant initial left and right states U_l and U_r , respectively. Note that this system can be rewritten in terms of the velocities $q = \hat{u}$ and $\hat{v}$ in the normal and tangent direction, as follows,

$$\begin{aligned} \partial_t h + \partial_{\hat{x}}(h q) &= 0 \\ \partial_t q + q \partial_{\hat{x}} q + \partial_{\hat{x}} h &= 0 \\ \partial_t \hat{v} + q \partial_{\hat{x}} \hat{v} &= 0. \end{aligned} \quad (4.3)$$

These are the symmetric shallow water equations in the $\hat{x}$ -direction, *i.e.* without $\hat{y}$ -dependence.

We are seeking an expression for $\hat{F} - V U_h$ on either side of a moving space-time cell edge, which (normal) velocity is denoted by V . In the Riemann problem we can distinguish four situations without drying or vacuum when either $P_l \neq 0, P_r \neq 0$ or $u_l + c_l > u_r + c_r$ with $c = \int^h a(w)/w dw$ and gravity-wave speed $a = \sqrt{\partial P / \partial h}$, and three more situations when drying is possible (see, *e.g.*, Toro, 1999). Drying occurs where $h = 0$.

Without dry states (emerging) the HLLC approach assumes that there are four states U_l, U_l^*, U_r^*, U_r from left to right separated from another by jumps at $\hat{x}/\hat{t} = S_l, S_m, S_r$ with $\hat{t} = t - t_n$ (taking the origin $\mathcal{O}$ without loss of generality at the face). We integrate (4.2) over two control volumes $\mathcal{ODCE}$ and $\mathcal{ABE}$ in Fig. 1 around the cell edge V in turn for each of the four cases (i) $S_l < V, S_r, S_m > V$; (ii) $S_l < V, S_m \leq V, S_r > V$; (iii) $S_l, S_m, S_r < V$; and iv) $S_l, S_m, S_r > V$. Subsequently, we calculate $\hat{F} - V_{\pm} U_h$ along V as the average of the contribution on either side of the edge V . After using Gauss' theorem in space and time on (4.2), the HLLC flux $\hat{F}_V^{hllc}$ is obtained. Details can be found in Appendix B (see also Van der Vegt and Van der Ven, 2002). The result is

$$\begin{aligned} \hat{F}_V^{hllc}(U_l, U_r) &= \frac{1}{2 \Delta t} \left(\int_{S_-} (\hat{F} - V U_h) \bar{n}_k^- dl_- + \int_{S_+} (\hat{F} - V U_h) \bar{n}_k^+ dl_+ \right) \\ &= \frac{1}{2} \left\{ \hat{F}_l + \hat{F}_r - (|S_l - V| - |S_m - V|) U_l^* + (|S_r - V| - |S_m - V|) U_r^* + \right. \\ &\quad \left. |S_l - V| U_l - |S_r - V| U_r - V (U_l + U_r) \right\}, \end{aligned} \quad (4.4)$$

where $\hat{F}_{l,r} = \hat{F}(U_{l,r})$, $\bar{n}_k^{\pm}$ are outward pointing normal vectors in space time, and $dl_{\pm}$ are pieces along the boundary $S_{\pm}$ of the area left and right of the contact interface at $S_m \hat{t}$. The intermediate states $U_{l,r}^*$ and speed S_m follow (Batten *et al.*, 1997) by a) using the intermediate one star state, $\tilde{U}^*$ emerging from the HLL approach, and b) using the Rankine-Hugoniot relations:

Ad. a) Using the HLL approach with one intermediate state $\tilde{U}^*$ to determine S_m , and assuming that $q_l^* = q_r^* = \tilde{q}^* = S_m$, we find that

$$\tilde{u}^* = S_m = \frac{P_l - P_r + h_r q_r (S_r - q_r) - h_l q_l (S_l - q_l)}{h_r (S_r - q_r) - h_l (S_l - q_l)} \quad (4.5)$$

and it may be checked that $P^* = P_l^* = P_r^* = P_l + h_l (S_l - q_l)/(S_m - q_l)$ as it should be for contact discontinuities with speed S_m . By retracing the derivation for S_m we find that $S_m = S_r = S_l$ when $h_r (S_r - q_r) - h_l (S_l - q_l) = 0$. The latter can happen when a dry patch is about to emerge.

Ad. b) By using the Rankine-Hugoniot relations for $\partial_t U + \partial_{\hat{x}} \hat{F} = 0$ we find

$$(S_{l,r} - S_m) U_{l,r}^* = (S_{l,r} - q_{l,r}) U_{l,r} + \begin{pmatrix} 0 \\ (P^* - P_{l,r}) \hat{n}_k \end{pmatrix} \quad (4.6)$$

with

$$P^* = P_{l,r}^* = P_{l,r} + h_{l,r} (S_{l,r} - q_{l,r}) (S_m - q_{l,r}) \quad (4.7)$$

and velocities $q_{l,r}$. By combining (4.4) and (4.6) we can deal with the cases when $S_{l,r} = S_m$. The latter handling is important when $h = 0$ at one of the faces on an element.

Finally, the wave speeds are estimated based on the left and right moving rarefaction wave speeds

$$S_l = \min(q_l - a_l, q_r - a_r), \quad S_r = \max(q_l + a_l, q_r + a_r), \quad (4.8)$$

respectively, with $a^2 = \partial P / \partial h$. When $S_l > V$ the flux simplifies to $\hat{F}_l$ and when $S_r < V$ to $\hat{F}_r$, *i.e.* the upwind cases.

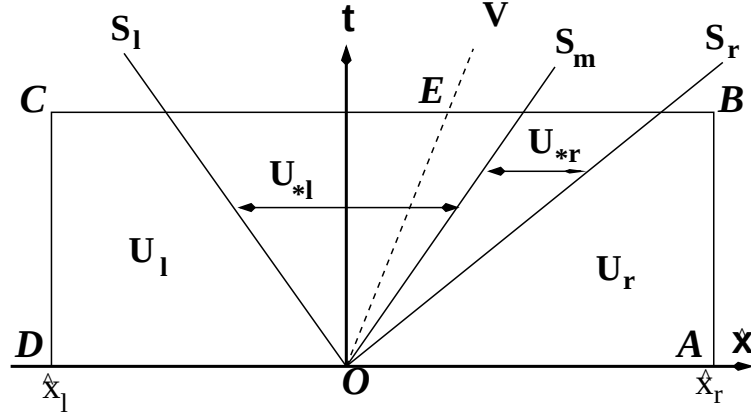


Figure 1: The wave configuration is sketched to define the variables and states involved in the HLLC flux approximation. The constant left and right states $U_{l,r}$ are separated by a left and right wave $S_{l,r}$ from two constant star regions $U_{*,l,r}$. These two star regions are divided by a contact wave S_m .

5 Flooding and Drying

5.1 Free boundary or edge elements

The tessellation $\mathcal{T}_{rh}$ of Ω consists of flooded and dry elements separated by allowable free-boundary paths. Generally, the depth h in the shallow-water equations is zero along the free boundary with the initial condition being the possible exception.

On triangular elements the depth is approximated by a mean and two slopes as follows:

$$h = \bar{h} + (\zeta_1 - 1/3) \hat{h}_1 + (\zeta_2 - 1/3) \hat{h}_2. \quad (5.1)$$

Within this approximation a nontrivial depth field at the free boundary, where $h = 0$, occurs only for triangular elements of either Type I with one fixed node and two free boundary nodes or, of type II with two fixed nodes and one free boundary node.

For Type I, either node a) $i = 0$, b) $i = 1$ or c) $i = 2$ is fixed, and if $h = 0$ at this node then

$$h = 3 \bar{h} \zeta_i \quad (5.2)$$

with $\zeta_0 = 1 - \zeta_1 - \zeta_2$. Similar constrained expansions are obtained for m, μ . Consequently, the velocity is constant in an element of type I and equals $(\bar{u}, \bar{v}) = (\bar{m}, \bar{\mu})/\bar{h}$.

For Type II, either nodes a) 0, 1, b) 0, 2 or c) 1, 2 are fixed. For case IIa), the free boundary is at location $\bar{x}_2$ where the depth $h = 0$; hence,

$$h = \frac{3}{2} \bar{h} (1 - \zeta_2) + \hat{h}_1 (\zeta_1 - (1 - \zeta_2)/2) \quad (5.3)$$

with $-3\bar{h} \leq \hat{h}_2 \leq 0$, $|\hat{h}_1| \leq 3\bar{h}$. For case IIb), the free boundary is at location $\bar{x}_1$; hence,

$$h = \frac{3}{2} \bar{h} (1 - \zeta_1) + \hat{h}_2 (\zeta_2 - (1 - \zeta_1)/2) \quad (5.4)$$

with $-3\bar{h} \leq \hat{h}_1 \leq 0$, $|\hat{h}_2| \leq 3\bar{h}$. For case IIc), the free boundary is at location $\bar{x}_0$; hence,

$$h = \hat{h}_1 \zeta_1 + (3\bar{h} - \hat{h}_1) \zeta_2 \quad \text{or} \quad h = (3\bar{h} - \hat{h}_2) \zeta_1 + \hat{h}_2 \zeta_2 \quad (5.5)$$

with $|\hat{h}_1|, |\hat{h}_2| \leq 3\bar{h}$. Similar constrained expansions are obtained for m, μ . The velocity at the free boundary in a type II free boundary element can therefore still vary, and can be determined at the free boundary node using l'Hôpital's rule.

5.2 Time-dependent tessellation topology

The triangulation process is as follows. The basis of the triangulation is a basic, fixed mesh of rectangles (which is not necessarily a rectangular mesh) overlaying the fixed computational domain. The mesh initialization entails three steps.

- i. First, the mesh is deformed locally near the computational boundary. Nearby nodes of the mesh are moved to lie on or near the fixed boundary, and diagonals of a deformed rectangle may form a close fit. This results in the basic fixed regular mesh of the computational domain consisting of rectangles in the interior, and quadrilaterals and triangles at the computational boundary.
- ii. Second, in a similar way the nodes of the basic fixed mesh of the domain are moved to lie on or near the initial free boundary, and diagonals may be placed to fit the initial free boundary approximately.
- iii. Third, the diagonals are placed in the remaining fixed flooded rectangles to complete the triangulation.

The free-boundary dynamics is incorporated by locally moving the nodes to the water line and updating the mesh:

- iv. The nodes on the free boundary are moved a limited amount such that the topology of the mesh is preserved during a time step. The preservation of the topology is ensured by the time-step restriction.
- v. After each time step, the basic grid of rectangles is deformed to fit the new position of the free boundary, which may imply that nodes formerly on the free boundary snap back to their location on the fixed grid, and, vice versa, formerly fixed nodes are moved to become free-boundary nodes. The free boundary may be smoothened during this process given the available resolution.
- vi. Finally, the triangulation is completed by cunningly placing or swapping diagonals in the deformed rectangles. Edge swapping is only used to swap the diagonal in a rectangle in order to define fixed and free boundary fits within the existing topology of the grid of rectangles.

The mesh initialization steps i–vi are detailed further by defining exactly what is meant by the movement of the free-boundary nodes, the deformation of nodes to a boundary, and the placement of diagonals in deformed rectangles.

5.2.1 Deformation of nodes to the initial boundaries

It is assumed that the computational boundary and the initial free boundary are sufficiently smooth such that these boundaries do not intersect the sides of a rectangle more than four times. The total free boundary is a union of free closed or open boundary segments: such a segment is either closed or begins or ends at different points of the computational boundary. Moreover, if there are two intersection points of the boundary with a rectangular edge these two nodes are not both closer to one node of this rectangular edge. Otherwise, in the latter case these nodes are collapsed as part of a general smoothing of the boundaries.

Consider the intersection points of the basic grid lines with the computational boundary or a free boundary segment. The node closest to an intersection point is moved along an edge of the rectangle to the intersection point, unless there are two intersection points concerned, in which case the node is moved along the (relevant) diagonal to the intersection point of the (free or computational) boundary with this diagonal. The approximated boundary is defined by connecting the moved boundary nodes, either by the edges of the deformed rectangles or by placing diagonals in a rectangle.

In this way first the basic computational mesh is defined and subsequently this basic mesh is deformed to fit the existing free boundaries. The intersection of the free boundary with the computational boundary is a special situation. We consider the special case for which the rectangular domain is a couple of connected rectangular mesh blocks, which allows us to move the boundary nodes along the straight stretches of boundary. In this special case open boundaries are straight.

5.2.2 Movement of free-boundary nodes

The movement of the free-boundary nodes is given by assigning an advection velocity to $\hat{x}_i$ with i the relevant local node number. Since the velocity is discontinuous at a free-boundary node, the mean of the values of the velocities in the surrounding flooded triangles is taken.

5.2.3 Updating the grid by deformation and edge swapping

Once the free-boundary nodes have moved an update of the mesh can be necessary. The flow variables are then defined on the existing pre-updated mesh. Each free boundary segment consists of connected straight lines, which intersects the underlying basic computational mesh at the edge of some of the rectangular elements. A new deformed mesh is thus defined using the same strategy of assigning nodes to the computational boundary and the initial free boundary. Since the computed free boundary has a finite resolution, smoothing is only necessary when pieces of one free boundary are moving to an intersection, or if two free-boundary segments move to an encounter, in which case the topology of the free boundary is changed. This topology can change also with the creation of a new free boundary in a flooded area. The old and new mesh need not coincide and through projection the variables are defined on the new mesh. Subsequently, the next time step can proceed.

5.3 Time-step criterion

To ensure preservation of the topology during a time step, a time-step criterion is presented in the following theorem.

Proposition 5.1. *Given the following assumptions and definitions:*

A1(a): *In flooded elements, the depth $h \geq 0$ and therefore the mean depth $\bar{h}_k \neq 0$. Assume that the depth and the momenta $m = h u$, $\mu = h v$ are represented by a linear polynomial consisting of a mean and two slopes on a triangle, as follows:*

$$\begin{aligned} h^n &= \bar{h}_k^n + (\zeta_1 - 1/3) \hat{h}_{1k}^n + (\zeta_2 - 1/3) \hat{h}_{2k}^n \\ m^n &= \bar{m}_k^n + (\zeta_1 - 1/3) \hat{m}_{1k}^n + (\zeta_2 - 1/3) \hat{m}_{2k}^n, \\ \mu^n &= \bar{\mu}_k^n + (\zeta_1 - 1/3) \hat{\mu}_{1k}^n + (\zeta_2 - 1/3) \hat{\mu}_{2k}^n, \end{aligned} \tag{5.6}$$

written in terms of reference coordinates ζ_1 and ζ_2 at time t^n . Furthermore, take momentum $m = h u$ such that $m = 0$ when $h = 0$.

A1(b): Consider the Euler forward time step discretization and the space discontinuous Galerkin discretization for the mean depth in the general formulation (3.18)

$$|K_k^{n+1}| \bar{h}_k^{n+1} = \bar{h}_k^n |K_k^n| - \Delta t \sum_{i=0}^{N_{el}^k-1} \int_{S_i} \hat{F}_{Vh,i}^{hllc} d\Gamma, \quad (5.7)$$

where in element k the mean depth at time $t^{n+1} = t^n + \Delta t$ is $\bar{h}_k^{n+1}$, and at time t^n is $\bar{h}_k^n$; $\hat{F}_{Vh,i}^{hllc}$ is the mass flux $(F \cdot \hat{n}_k - V U)$ at face S_i and time t^n of an element and $d\Gamma$ an infinitesimal distance coordinate along a face; and N_{el}^k is the number of nodes and also the number of faces for element k .

A1(c): Assume the HLLC flux scheme (4.4)–(4.8) is used for the flux $\hat{F}_h$ in (5.7).

A1(d): Calculated finite velocities $u = m/h$ using l'Hopital's rule when $h = 0$ since then $m = 0$ (assumption 1(a)), and approximate u for small h .

A1(e): The area $|K_k|$ at time t^n is positive.

Definition 1: Each interior and boundary face has an element to the left, k_{left} , and each interior face also has an element to the right, k_{right} .

Definition 2: Nodes of elements in the interior are fixed; triangular elements at the boundary have therefore either two (moving) free boundary node (Type I), or one (moving) free boundary node (Type II).

Given assumptions 1(a)–(e) and definitions 1-2, the mean depth $\bar{h}^{n+1}$ at the next time step is positive when the maximum of an inverse time step $\Delta\mu_k = |K_k|/\Delta t_k$ scaled by the element area $|K_k|$ is taken at each face.

By visiting all interior faces, the contributions to $\Delta\mu_{k_{left}, k_{right}}$ are estimated in the sum over the the quadrature points at each face, as follows:

for all elements

$$\Delta\mu_k = 0$$

for all interior faces

for all quadrature points per face

if $S_m - V > 0$ then

if $S_l - V < 0$ then

$$\Delta\mu_{k_{left}+} = 3 \frac{(S_m - V)(q_l - S_l) L_{face} w_q}{S_m - S_l}$$

else ($S_l - V \geq 0$)

$$\Delta\mu_{k_{left}+} = 3 (q_l - V) L_{face} w_q$$

else ($S_m - V \leq 0$)

if $S_r - V < 0$ then

$$\Delta\mu_{k_{right}+} = 3 |q_r - V| L_{face} w_q$$

else ($S_r - V \geq 0$)

$$\Delta\mu_{k_{right}+} = 3 \frac{|S_m - V| (S_r - q_r) L_{face} w_q}{S_r - S_m}, \quad (5.8)$$

where L_{face} is the length of a face, w_q the weight for the quadrature point, and the (c-language) convention is used such that $a+ = 1$ means $a = a + 1$. Finally, by visiting all wet faces of the computational boundary, the contributions to $\Delta\mu_{k_{left}, k_{right}}$ are estimated in the sum over the the

quadrature points at each face, as follows:

for all wet faces of the computational boundary

for all quadrature points per face

if $S_m > 0$ then

if $S_l < 0$ then

$$\Delta \mu_{k_{left}} = 3 \frac{S_m (q_l - S_l) L_{face} w_q}{S_m - S_l}$$

else ($S_l \geq 0$)

$$\Delta \mu_{k_{left}} = 3 q_l L_{face} w_q. \quad (5.9)$$

For the second-order Gauss quadrature $w_q = 1$ (see Appendix A). All terms on the right-hand-side in (5.8)–(5.9) are evaluated at time t^n .

Proof. First, the HLLC flux $\hat{F}_h^{hllc}$ is considered. Define the shorthand $S_{lV} = S_l - V$, etcetera, where V is the normal velocity of the face (at a quadrature point). The line integrals are approximated using (Gauss) quadrature, *i.e.*,

$$\int_{S_i} \hat{F}_{V,h}^{hllc} d\Gamma = \sum_q w_q \hat{F}_{V,h}^{hllc}(\bar{x}_q) \quad (5.10)$$

with $\bar{x}_q$ the coordinate of the relevant quadrature point along face S_i . The subscript in $\hat{F}_{V,h}^{hllc}$ denotes that we consider the depth component h of the flux (4.4), *i.e.*

$$\begin{aligned} 2 \hat{F}_{V,h}^{hllc} &= q_l h_l + q_r h_r - (|S_{lV}| - |S_{mV}|) h_l^* + (|S_{rV}| - |S_{mV}|) h_r^* + \\ &\quad |S_{lV}| h_l - |S_{rV}| h_r - V (h_l + h_r) \\ &= ((S_l - S_m + |S_{lV}| - |S_{mV}|) q_l + (|S_{mV}| - V) S_l + (V - |S_{lV}|) S_m) \frac{h_l}{S_l - S_m} + \\ &\quad ((S_r - S_m - |S_{rV}| + |S_{mV}|) q_r - (|S_{mV}| + V) S_r + (V + |S_{rV}|) S_m) \frac{h_r}{S_r - S_m}. \end{aligned} \quad (5.11)$$

The flux (5.11) at a face reduces to the following four cases

$$\hat{F}_{V,h}^{hllc} = \begin{cases} h_r (q_r - V) < 0 & \text{if } S_{lV} < S_{mV} < S_{rV} < 0 \\ h_r (V - S_m) (q_r - S_r) / (S_r - S_m) < 0 & \text{if } S_{lV} < S_{mV} < 0 \wedge S_{rV} > 0 \\ h_l (q_l - V) > 0 & \text{if } 0 < S_{lV} < S_{mV} < S_{rV} \\ h_l (S_m - V) (q_l - S_l) / (S_m - S_l) > 0 & \text{if } S_{lV} < 0 \wedge 0 < S_{mV} < S_{rV} \end{cases}, \quad (5.12)$$

in which we used relation (4.8) to determine the inequalities.

Second, the contributions due to the time dependence of the elements are considered. For the triangular free-boundary or edge elements the minimum value of h at the nodes is zero by assumption 1 and a linear polynomial representation (5.6) is used for h, m, μ . Distinguish triangles with two free-boundary nodes (Type I), and ones with one free-boundary node (Type II). For Type I, $h = 0$ at two free-boundary nodes and $h = 3 \bar{h} \zeta_i$ with cases Ia) $i = 0$, Ib) $i = 1$, and Ic) $i = 2$ the local nonzero third node and $\zeta_0 = 1 - \zeta_1 - \zeta_2$. Hence, for type I we have:

$$\begin{aligned} \dot{x}_0 = 0 \quad \hat{h}_1 = \hat{h}_2 = -3 \bar{h} & \quad \text{for case Ia,} \\ \dot{x}_1 = 0 \quad \hat{h}_1 = 3 \bar{h}, \hat{h}_2 = 0 & \quad \text{for case Ib,} \\ \dot{x}_2 = 0 \quad \hat{h}_1 = 0, \hat{h}_2 = 3 \bar{h} & \quad \text{for case Ic.} \end{aligned} \quad (5.13)$$

For Type II, $h = 0$ only at the free boundary node, and hence:

$$\begin{aligned} h &= \frac{3}{2} \bar{h} (1 - \zeta_2) + \hat{h}_1 (\zeta_1 - (1 - \zeta_2)/2) \quad \text{for case IIa } \dot{x}_0 = \dot{x}_1 = 0 \\ h &= \frac{3}{2} \bar{h} (1 - \zeta_1) + \hat{h}_2 (\zeta_2 - (1 - \zeta_1)/2) \quad \text{for case IIb } \dot{x}_0 = \dot{x}_2 = 0 \\ h &= 3 \bar{h} \zeta_1 + \hat{h}_2 (\zeta_2 - \zeta_1) \quad \text{for case IIc } \dot{x}_1 = \dot{x}_2 = 0. \end{aligned} \quad (5.14)$$

In all these cases for Type II, a non-negative depth implies $|\hat{h}| \leq 3\bar{h}$, which can be used to relate the slope information to the mean depth. To wit, for Type II:

$$\begin{aligned} \dot{x}_0 = \dot{x}_1 = 0 & \quad -3\bar{h} \leq \hat{h}_2 \leq 0 \text{ and } -3\bar{h} \leq \hat{h}_1 \leq 3\bar{h} & \text{for case IIa,} \\ \dot{x}_0 = \dot{x}_2 = 0 & \quad -3\bar{h} \leq \hat{h}_1 \leq 0 \text{ and } -3\bar{h} \leq \hat{h}_2 \leq 3\bar{h} & \text{for case IIb,} \\ \dot{x}_1 = \dot{x}_2 = 0 & \quad 0 \leq \hat{h}_1, \hat{h}_2 \leq 3\bar{h} & \text{for case IIc.} \end{aligned} \quad (5.15)$$

Observe that the HLLC depth flux (5.12) is linear in h , and that the contribution due to the element time dependence is linear in the slopes $\hat{h}$. Combining the results in (5.12), (5.13), and (5.15), with worst-case estimates expressing $\hat{h}_1^n$ and $\hat{h}_2^n$ into inequalities linear in $\bar{h}^n$, the discretized mean depth equation (5.7) becomes a linear inequality in $\bar{h}^n$. Using a maximum principle (*e.g.*, Morton and Mayers, 1994), the largest of the intermediate time steps found in (5.8)–(5.9) guarantees positivity of the mean depth $\bar{h}^{n+1}$. $\square$

Remarks:

- (i) The proposition extends to the third-order Runge-Kutta scheme by repeating the proposition for every time stage h^i , assuming subsequently that $h^i \geq 0$ and $\bar{h}^i > 0$. Unfortunately, a smaller time step at later (intermediate) time steps may require to start again at the beginning.
- (ii) The proposition is valid for any continuity equation using the same discretization. Hence, it is not restricted to the shallow-water case, but also valid, for example, for the compressible Euler equations by replacing h by a suitable (pseudo) density σ .
- (iii) The extension of the proposition from one dimension (Bokhove, 2003) to two dimensions suggests that extensions to higher-order polynomial basis functions or higher dimensions are similar in spirit but more involved.
- (iv) The proof shows that the minimum time step per element is most easily implemented numerically in parallel with the loop over the faces to determine the fluxes. Hence, the time step should be set after the fluxes and time-step criterion have been determined in unison.
- (v) The conditions in remark (i) are ensured by limiting the slopes of h and in parallel m and μ to zero at one or two nodes of the triangular element whenever at intermediate stages $h^i < 0$ (but $\bar{h}^i > 0$ by Proposition 5.1).

5.4 Slope limiter for drying events

While the mean depth in an element can be guaranteed to be positive subject to the time-step restriction in Proposition 5.1, the slopes can increase to cause the water depth at an interior flooded triangle to become negative. Given $\bar{h} > 0$, this may happen at a certain time stage. Such negative undershoots are fixed after each time stage by shifting the straight water line to the closest corner point. After this slope-limiting process the water line touches at most one node or coincides with an edge. Hence the slope-limited depth is non-negative in each triangle.

To wit, before the adjustment

$$h = \bar{h} + (\zeta_1 - 1/3)\bar{h}_1 + (\zeta - 1/3)\bar{h}_2 \quad (5.16)$$

on a triangle. Per triangle, the smallest value of the depth is found, and the relevant node number is stored. If this depth value is negative the adjusted depth depends on this node number:

$$h = \begin{cases} \frac{3\bar{h}}{\hat{h}_1 + \hat{h}_2} (\hat{h}_1 \zeta_1 + \hat{h}_2 \zeta_2) & \text{for node 0} \\ \frac{3\bar{h}}{-2\hat{h}_1 + \hat{h}_2} (\hat{h}_1 (\zeta_1 - 1) + \hat{h}_2 \zeta_2) & \text{for node 1} \\ \frac{3\bar{h}}{\hat{h}_1 - 2\hat{h}_2} (\hat{h}_1 \zeta_1 + \hat{h}_2 (\zeta_2 - 1)) & \text{for node 2} \end{cases} \quad (5.17)$$

6 Conclusions

A novel algorithm has been developed to deal with free-boundary dynamics in the two-dimensional shallow-water equations with a space discontinuous finite-element method. The numerical algorithm is able to preserve the non-negative nature of the (mean) depth in combination with a free-boundary treatment in mixed Eulerian-Lagrangian elements.

The numerical algorithm developed will be tested in future work by comparing the numerical solutions with various one- and two-dimensional exact solutions in which flooding and drying occurs on partially structured meshes: Riemann problems with drying (*e.g.*, Smoller, 1994; Toro, 1999), the one-dimensional Carrier-Greenspan (1958) solution and asymptotic two-dimensional version (Brocchini and Peregrine, 1996), the parabolic bowl solution (Young, 1986; Allen and Newberger, 1993), and the Peregrine-Williams (2001) solution.

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A Details of Finite-Element Discretization

A.1 Geometry and polynomial basis functions

In the basis element $\hat{K}$ we define basis functions

$$\hat{\varphi}_0(\bar{\zeta}) = 1, \quad \text{and} \quad \hat{\varphi}_m(\bar{\zeta}) = \zeta_1^{k_m-l} \zeta_2^l \quad \text{for} \quad l = 1, \dots, k_m \quad \text{and} \quad k_m \leq d_P \quad (\text{A.1})$$

with d_P the maximum degree of the polynomials used and $m = 1, \dots, M$ for the appropriate integer M . Finally we relate the local basis functions in $\hat{K}$ to the basis functions in K_k as follows

$$\hat{\varphi}_n(\bar{\zeta}) = \hat{\varphi}_n(F_{K_k}^{-1}(\bar{x}, t)) = \varphi_{n,k}(\bar{x}, t) \quad \text{for} \quad n = 0, \dots, d_P. \quad (\text{A.2})$$

In principle elements can be time dependent by allowing the nodes to move in time.

The variables U and test functions v are approximated in each element K_k by their polynomial approximations U_h and v_h , respectively, as follows

$$U_h(\bar{x}, t) = \sum_{m=0}^M \hat{U}_m(K_k, t) \psi_m(\bar{x}, t) \quad \text{and} \quad v_h(\bar{x}, t) = \sum_{m=0}^M \hat{v}_m(K_k) \psi_m(\bar{x}, t) \quad (\text{A.3})$$

with polynomial basis functions $\psi_m(\bar{x}, t)$'s $\in P^{d_P}(K_k)$. These are chosen such that $\hat{U}_{0k} = \bar{U}_k$ represents the mean and $\hat{U}_{mk}$ with $m > 0$ the higher-order projections of U on U_h , as follows

$$\begin{aligned} \hat{U}_0 &= \bar{U}(K_k, t) = \int_{K_k} U(\bar{x}, t) d\bar{x} / |K_k| \quad \text{and} \\ \psi_{m,k}(\bar{x}, t) &= \begin{cases} 1 & \text{if } m = 0 \\ \varphi_{m,k}(\bar{x}, t) - c_{m,k} & \text{if } m \geq 1 \end{cases} \end{aligned} \quad (\text{A.4})$$

with

$$c_{m,k} = \int_{K_k} \varphi_{m,k}(\bar{x}, t) d\bar{x} / |K_k|. \quad (\text{A.5})$$

For triangles, $c_{m,k} = 1/3$.

A.2 Normal vectors and transformations

The three normal vectors for triangular elements are

$$\begin{aligned} \hat{n}_0 &= \frac{1}{|\bar{x}_1 - \bar{x}_0|} (y_1 - y_0, x_0 - x_1)^T, & \hat{n}_1 &= \frac{1}{|\bar{x}_2 - \bar{x}_1|} (y_2 - y_1, x_1 - x_2)^T, \\ \hat{n}_2 &= \frac{1}{|\bar{x}_0 - \bar{x}_2|} (y_0 - y_2, x_2 - x_0)^T. \end{aligned} \quad (\text{A.6})$$

The Jacobian J_3 of the transformation between $\bar{x}$ and $\bar{\zeta}$ is

$$J_3 = \begin{pmatrix} \partial_{\zeta_1} x & \partial_{\zeta_1} y \\ \partial_{\zeta_2} x & \partial_{\zeta_2} y \end{pmatrix} = \begin{pmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \end{pmatrix}. \quad (\text{A.7})$$

Hence, gradients in $\bar{x}$ transform to

$$\begin{pmatrix} \partial_x V \\ \partial_y V \end{pmatrix} = \frac{1}{\det J_3} \begin{pmatrix} y_2 - y_0 & y_0 - y_1 \\ x_0 - x_2 & x_1 - x_0 \end{pmatrix} \begin{pmatrix} \partial_{\zeta_1} V \\ \partial_{\zeta_2} V \end{pmatrix} \quad (\text{A.8})$$

with determinant $\det J_3$ and $|J_3| = |\det J_3|$. Triangles have three sides $S_1, \dots, S_3$ spanned by node pairs $(\bar{x}_0, \bar{x}_1), \dots, (\bar{x}_2, \bar{x}_0)$. The sides $S_0, \dots, S_2$ correspond to $\zeta_2 = 0, \zeta_1 + \zeta_2 = 1$ and $\zeta_1 = 0$ in the reference element, respectively.

The absolute value of the determinant of these Jacobians is written as $|J|$. For interior and boundary integrals, we use

$$dK = |J| d\bar{\zeta}, \quad d\Gamma = |\bar{x}(l)| dl, \quad (\text{A.9})$$

respectively, with dl a cord on an element side and l an appropriate local coordinate. For triangles, we find

$$\begin{aligned} \int_{S_0} f^{tn}(\bar{x}) d\Gamma &= \int_0^1 f^{tn}(\zeta_1, 0) |\bar{x}_1 - \bar{x}_0| d\zeta_1 \quad \text{and} \quad \zeta_2 = 0, \\ \int_{S_1} f^{tn}(\bar{x}) d\Gamma &= \int_0^1 f^{tn}(1 - \zeta_2, \zeta_2) |\bar{x}_2 - \bar{x}_1| d\zeta_2 \quad \text{and} \quad \zeta_1 = 1 - \zeta_2, \\ \int_{S_2} f^{tn}(\bar{x}) d\Gamma &= \int_0^1 f^{tn}(0, \zeta_2) |\bar{x}_0 - \bar{x}_2| d\zeta_2 \quad \text{and} \quad \zeta_1 = 0. \end{aligned} \quad (\text{A.10})$$

A.3 Elemental integrals

The system of ordinary differential equations under consideration is (3.16) for $M = 2$

$$\begin{aligned} |K_k| \frac{d}{dt} \bar{U}_k + \hat{U}_{k,n} \int_{K_k} \partial_t \psi_{n,k} dK - \int_{K_k} S dK + \int_{\partial K_k} \tilde{F} \cdot \hat{n}_k d\Gamma &= 0 \\ \frac{d\hat{U}_{k,n}}{dt} \int_{K_k} \psi_{m,k} \psi_{n,k} dK + \hat{U}_{k,n} \int_{K_k} \psi_{m,k} \partial_t \psi_{n,k} dK - \int_{K_k} \psi_{m,k} S dK - \int_{K_k} F \cdot \text{grad} \psi_{m,k} dK + \\ \hat{U}_{k,n} \int_{K_k} (\text{grad} \psi_{m,k})^T \cdot D(U_h) : (\text{grad} \psi_{n,k})^T dK + \int_{\partial K_k} \psi_{m,k}^- \tilde{F} \cdot \hat{n}_k d\Gamma &= 0 \end{aligned} \quad (\text{A.11})$$

with $m, n = 1, \dots, M$. For triangles we use $\phi_m = 1, \zeta_1, \zeta_2, M = 2$.

For triangles, the area of an element is

$$|K_k| = \int_{K_k} dK = \int_0^1 \int_0^{1-\zeta_2} |J_3| d\zeta_1 d\zeta_2 = \frac{1}{2} |J_3|. \quad (\text{A.12})$$

Given the transformations of the previous sections in this appendix, the integrals in (A.11) can be evaluated in the reference coordinate system. For the integrals resulting from moving nodes, we have

$$E_{mn} = \int_{K_k} \psi_{m,k} \partial_t \psi_{n,k} dK \quad (\text{A.13})$$

for $m = 0, 1, 2$ and $n = 1, 2$. For triangles

$$E_{0n} = - \int_{\partial K} V \cdot \hat{n}_k \psi_{n,k} d\Gamma$$

with V the velocity vector of the relevant face due to the movement of the two connected nodes. In particular, we have

$$\begin{aligned}
E_{01} &= \frac{\text{sign}(J_3)}{6} \left((y_0 - y_2) (\dot{x}_1 - \dot{x}_0) + (x_2 - x_0) (\dot{y}_1 - \dot{y}_0) + (y_0 - y_2) (\dot{x}_2 - \dot{x}_0) + \right. \\
&\quad \left. (x_2 - x_0) (\dot{y}_2 - \dot{y}_0) + 3 (y_0 - y_2) \dot{x}_0 + 3 (x_2 - x_0) \dot{y}_0 \right) \\
E_{02} &= \frac{\text{sign}(J_3)}{6} \left((y_1 - y_0) (\dot{x}_2 - \dot{x}_0) + (y_1 - y_0) (\dot{x}_1 - \dot{x}_0) + (x_0 - x_1) (\dot{y}_1 - \dot{y}_0) + \right. \\
&\quad \left. (x_0 - x_1) (\dot{y}_2 - \dot{y}_0) + 3 (y_1 - y_0) \dot{x}_0 + 3 (x_0 - x_1) \dot{y}_0 \right) \\
E_{11} &= \frac{\text{sign}(J_3)}{72} \left(2 (y_0 - y_2) (\dot{x}_1 - \dot{x}_0) + 2 (x_2 - x_0) (\dot{y}_1 - \dot{y}_0) + (y_2 - y_0) (\dot{x}_2 - \dot{x}_0) + \right. \\
&\quad \left. (x_0 - x_2) (\dot{y}_2 - \dot{y}_0) \right) \\
E_{21} &= \frac{\text{sign}(J_3)}{72} \left((y_2 - y_0) (\dot{x}_1 - \dot{x}_0) + (x_0 - x_2) (\dot{y}_1 - \dot{y}_0) + 2 (y_0 - y_2) (\dot{x}_2 - \dot{x}_0) + \right. \\
&\quad \left. 2 (x_2 - x_0) (\dot{y}_2 - \dot{y}_0) \right) \\
E_{12} &= \frac{\text{sign}(J_3)}{72} \left(2 (y_1 - y_0) (\dot{x}_1 - \dot{x}_0) + 2 (x_0 - x_1) (\dot{y}_1 - \dot{y}_0) + (y_0 - y_1) (\dot{x}_2 - \dot{x}_0) + \right. \\
&\quad \left. (x_1 - x_0) (\dot{y}_2 - \dot{y}_0) \right) \\
E_{22} &= \frac{\text{sign}(J_3)}{72} \left((y_0 - y_1) (\dot{x}_1 - \dot{x}_0) + (x_1 - x_0) (\dot{y}_1 - \dot{y}_0) + 2 (y_1 - y_0) (\dot{x}_2 - \dot{x}_0) + \right. \\
&\quad \left. 2 (x_0 - x_1) (\dot{y}_2 - \dot{y}_0) \right)
\end{aligned} \tag{A.14}$$

with $\dot{x}_0 = dx_0/dt$, etcetera.

To use the same flux $F_V = \hat{F} - V U$ in the mean and fluctuation equation, the additional time dependent term is combined with the flux due to the movement of the grid as follows

$$\begin{aligned}
E_{mn} \hat{U}_n + \sum_{S_i} \psi_{m,k}^- (\bar{U}_k + \psi_n \hat{U}_k) V d\Gamma &= \int_{K_k} \psi_{m,k} \partial_t \psi_{n,k} dK \hat{U}_n + \int_{K_k} \partial_{x_j} (V_j \psi_m) U_k dK \\
&= G_m (U_k).
\end{aligned} \tag{A.15}$$

For the integrals of the external contributions over an element interior with $m = 0, 1, 2$, we find

$$S_m = \int_{K_k} \psi_{m,k} S(\bar{x}) dK = \int_0^1 \int_0^{1-\zeta_2} (\varphi_m - c_{m,k}) S(\bar{\zeta}) |J_3| d\zeta_1 d\zeta_2. \tag{A.16}$$

For the topographic part of S , i.e., $S_b = (0, -h \partial_x h_b, -h \partial_y h_b)^T$, it is crucial that the pressure components of the flux balance the topographic terms in case of a flow at rest. Hence, for triangles and smooth topography we take $h_b = h_{b0} (1 - \zeta_1 - \zeta_2) + h_{b1} \zeta_1 + h_{b2} \zeta_2$ with $h_{b0,1,2}$ the values of the topography at the nodes and $\phi_m = \zeta_1, \zeta_2$. Hence,

$$S_{bm} = - \int_0^1 \int_0^{1-\zeta_2} (\varphi_m - c_{m,k}) h \begin{pmatrix} 0 \\ \partial_{\zeta_2} y \partial_{\zeta_1} h_b - \partial_{\zeta_1} y \partial_{\zeta_2} h_b \\ -\partial_{\zeta_2} x \partial_{\zeta_1} h_b + \partial_{\zeta_1} x \partial_{\zeta_2} h_b \end{pmatrix} \text{sign}(J_3) d\zeta_1 d\zeta_2. \tag{A.17}$$

For the mass matrix M_{mn} with $m, n = 0, 1, 2$

$$M_{mn} = \int_{K_k} \psi_{m,k} \psi_{n,k} dK = \begin{cases} |K_k| & \text{for } m = n = 0 \\ 0 & \text{for } m = 0, n \neq 0 \text{ or } n = 0, m \neq 0 \end{cases} \tag{A.18}$$

and for $m, n \neq 0$

$$\mathbf{M}_{mn} = \int_{K_k} \psi_{m,k} \psi_{n,k} dK = \int_0^1 \int_0^{1-\zeta_2} (\varphi_m - c_{m,k}) (\varphi_n - c_{n,k}) |J_3| d\zeta_1 d\zeta_2. \quad (\text{A.19})$$

For the interior flux term with $m = 1, 2$, for triangles

$$\mathbf{F}_m = \int_{K_k} F \cdot \text{grad } \psi_{m,k} dK = \int_0^1 \int_0^{1-\zeta_2} F \cdot \begin{pmatrix} (y_2 - y_0) \delta_{1m} + (y_0 - y_1) \delta_{2m} \\ (x_0 - x_2) \delta_{1m} + (x_1 - x_0) \delta_{2m} \end{pmatrix} \text{sign}(J_3) d\zeta_1 d\zeta_2. \quad (\text{A.20})$$

The dissipation matrix is

$$\mathbf{D}_{mn} = \int_{K_k} (\text{grad } \psi_{m,k})^T \cdot D(U_h) : (\text{grad } \psi_{n,k})^T dK \quad (\text{A.21})$$

with $m, n = 1, 2$.

To evaluate these interior integrals numerically second-order Gauss quadrature is used. For triangles

$$\int_0^1 \int_0^{1-\zeta_2} f^{tn}(\zeta_1, \zeta_2) d\zeta_1 d\zeta_2 \approx \frac{1}{6} \left(f^{tn}(0, c_t) + f^{tn}(-c_t, 0) + f^{tn}(c_t, c_t) \right) \quad (\text{A.22})$$

with $c_t = 1/2$.

The integration over element boundaries is performed by collecting the integration of

$$\int_S (\psi_m^l - \psi_m^r) \tilde{F}(U_l, U_r, \hat{n}) \cdot \hat{n}^l d\Gamma \quad (\text{A.23})$$

over all faces and assigning these to the respective elements. All faces either belong to the set of internal faces or the set of boundary faces for which $v_h^r = 0$. A counterclockwise convention is used for the integration along the faces: for the element left of a face the node number therefore increases (cyclically) and vice versa for the element right of the same face. The list of faces consists of the left and right element number and the local face number of these left and right elements. For boundary faces the type of boundary is additional with extra external information for certain boundary conditions such as inflow specification. The rest of the information is then available from the element list, such as the element type (triangular), the coordinates of the nodes, information on refinement or coarsening at certain faces, and links to element neighbors.

The flux terms are always evaluated by integration in the interior, left element since $\tilde{F}(U_l, U_r, \hat{n}) \cdot \hat{n}^l$ is continuous in the left and right element. Only the values of ψ_m^r and U_r need to be found. Assigning signs $s_{l,r} = +1$ for sides $S_{1,2}$ and $s_{l,r} = -1$ for sides S_3 to the left and right elements for triangles, to values $\zeta_{li} = \pm c_w$ or $(1 \pm c_w)/2$ on the left correspond values $\zeta_{rj} = \mp s_l s_r c_w$ or $\zeta_{rj} = (1 \mp s_l s_r c_w)/2$ on the right, in which the local coordinates in the left and right element are denoted by $\tilde{\zeta}_l$ and $\tilde{\zeta}_r$, respectively constrained at the corresponding faces. These resulting relations follow by putting each possible face of a triangle of the left element along each possible face of a triangle of the right element.

The product of $\hat{n}_k d\Gamma$ in the flux integrals over the boundaries simplify. For triangles we find

$$\begin{aligned} \mathbf{B}_{m0} &= \int_{S_0} \psi_{m,k}^- \hat{F}_V \cdot \hat{n}_k d\Gamma = \int_0^1 \psi_{m,k}^- \hat{F}_V \cdot (y_1 - y_0, x_0 - x_1)^T d\zeta_1 \quad \text{and} \quad \zeta_2 = 0, \\ \mathbf{B}_{m1} &= \int_{S_1} \psi_{m,k}^- \hat{F}_V \cdot \hat{n}_k d\Gamma = \int_0^1 \psi_{m,k}^- \hat{F}_V \cdot (y_2 - y_1, x_1 - x_2)^T d\zeta_2 \quad \text{and} \quad \zeta_1 = 1 - \zeta_2, \\ \mathbf{B}_{m2} &= \int_{S_2} \psi_{m,k}^- \hat{F}_V \cdot \hat{n}_k d\Gamma = \int_0^1 \psi_{m,k}^- \hat{F}_V \cdot (y_0 - y_2, x_2 - x_0)^T d\zeta_2 \quad \text{and} \quad \zeta_1 = 0. \end{aligned} \quad (\text{A.24})$$

Introducing these shorthand notations, the system of ordinary equations (A.11) reads

$$\begin{aligned} \frac{d(|K_k| \bar{U}_k)}{dt} - S_0 + \sum_{i=0}^{N_{el}^k-1} B_{0i} &= 0 \\ M_{mn} \frac{d\hat{U}_{k,n}}{dt} - S_m - F_m + G_m(U_k) + D_{mn} \hat{U}_{k,n} + \sum_{i=0}^{N_{el}^k-1} B_{mi} &= 0 \end{aligned} \quad (A.25)$$

with $m, n = 1, 2$ and all matrices depending on k .

B HLLC flux

Integrate (4.2) over the regions $OECD$ and $OABE$, in Fig. 1 and use Gauss' theorem in space time $\hat{x}-t$ to find

$$\begin{aligned} \int_{\hat{x}_l}^{S_l \Delta t} U_l d\hat{x} + \int_{S_l \Delta t}^V U_l^* d\hat{x} - \int_{\hat{x}_l}^O U_l d\hat{x} - \int_{t_n}^{t_n+\Delta t} \hat{F}_l dt + \int_{t_n}^{t_n+\Delta t} (\hat{F}_- - V U_-) dt &= 0 \\ \int_{V \Delta t}^{S_m \Delta t} U_l^* d\hat{x} + \int_{S_m \Delta t}^{S_r \Delta t} U_r^* d\hat{x} + \int_{t_n}^{t_n+\Delta t} \hat{F}_r dt - \int_O^{\hat{x}_r} U_r d\hat{x} + \int_{S_r \Delta t}^{\hat{x}_r} U_r d\hat{x} - \\ \int_{t_n}^{t_n+\Delta t} (\hat{F}_+ - V U_+) dt &= 0, \end{aligned} \quad (B.1)$$

respectively. Rewrite these results as

$$\begin{aligned} \int_{t_n}^{t_n+\Delta t} (\hat{F}_- - V U_-) dt &= (\hat{F}_l + U_l^* (S_l - V) - S_l U_l) \Delta t \\ \int_{t_n}^{t_n+\Delta t} (\hat{F}_+ - V U_+) dt &= (\hat{F}_r + U_l^* (S_m - V) + U_r^* (S_r - S_m) - S_r U_r) \Delta t. \end{aligned} \quad (B.2)$$

The HLLC flux (4.4) is the mean of the contributions $\hat{F}_\pm - V U_\pm$ in (B.2). The general expression (4.4) also covers the three other cases with (ii) $S_l < V, S_m \leq V, S_r > V$; (iii) $S_l, S_m, S_r < V$; and iv) $S_l, S_m, S_r > V$.

However, the unknowns $U_{l,r}^*, S_{l,r}$ and S_m still require determination. First, S_m is determined by assuming that $S_m = q_l^* = q_r^* = q^* = u_{HLL}^*$ equals the velocity in the HLL approach. In this approach, only one constant, star state U_{HLL}^* is introduced between $U_{l,r}$ as approximation. Assuming for the moment that $U^* = U_{HLL}^* = U_{l,r}^*$ (4.2) is integrated over $OABECD$ in Fig. 1, as follows

$$\int_{S_l \Delta t}^{S_r \Delta t} U^* d\hat{x} + \int_{t_n}^{t_n+\Delta t} \hat{F}_r dt - \int_O^{S_r \Delta t} U_r d\hat{x} - \int_{S_l \Delta t}^O U_l d\hat{x} - \int_{t_n}^{t_n+\Delta t} \hat{F}_l dt = 0. \quad (B.3)$$

Hence, from (B.3) we obtain

$$\begin{aligned} U^* &= \frac{S_r U_r - S_l U_l - (\hat{F}_r - \hat{F}_l)}{S_r - S_l} \quad \text{and} \\ S_m = q^* &= \frac{h_r q_r (S_r - q_r) - h_l q_l (S_l - q_l) + P_l - P_r}{h_r (S_r - q_r) - h_l (S_l - q_l)} \end{aligned} \quad (B.4)$$

by using $U = (h, h u, h v)^T$ and $q = (u, v)^T \cdot \hat{n}_k$. Second, the Rankine-Hugoniot relations around $\hat{x}_l = S_l \Delta t$ are

$$\begin{aligned} S_l U_l^* - \hat{F}_l^* &= S_l U_l - \hat{F}_l \quad \text{or} \\ \begin{cases} (S_l - q_l^*) h_l^* &= (S_l - q_l) h_l \\ (S_l - q_l^*) h_l^* u_l^* - P^* n_1 &= (S_l - q_l) h_l u_l - P_l n_1 \\ (S_l - q_l^*) h_l^* v_l^* - P^* n_2 &= (S_l - q_l) h_l v_l - P_l n_2. \end{cases} \end{aligned} \quad (B.5)$$

Taking the inner product of the last two relations with $\hat{n}_k$ gives

$$(S_l - q_l^*) h_l^* q_l^* + P^* = S_l h_l q_l - h_l q_l^2 - P_l. \quad (\text{B.6})$$

Third, since we know $q^* = S_m$ from (B.4), it follows from (B.5)a that

$$h_l^* = \frac{S_l - q_l}{S_l - S_m} h_l. \quad (\text{B.7})$$

Using $q_l^* = q^* = S_m$ in (B.6) yields

$$P^* = P_l + h_l (S_l - q_l) (S_m - q_l). \quad (\text{B.8})$$

As P^* and $S_m = q^*$ are known from (B.4) and (B.8) we can finally determine $h_l^* u_l^*$ and $h_l^* v_l^*$ from (B.5). Likewise, we can find U_r^* and, hence, find (4.6).

C Numerical Implementation

The numerical algorithm is implemented in the *c*-programming language.